Fields and Waves I

Lecture 11

Electric Potential & Electric Materials

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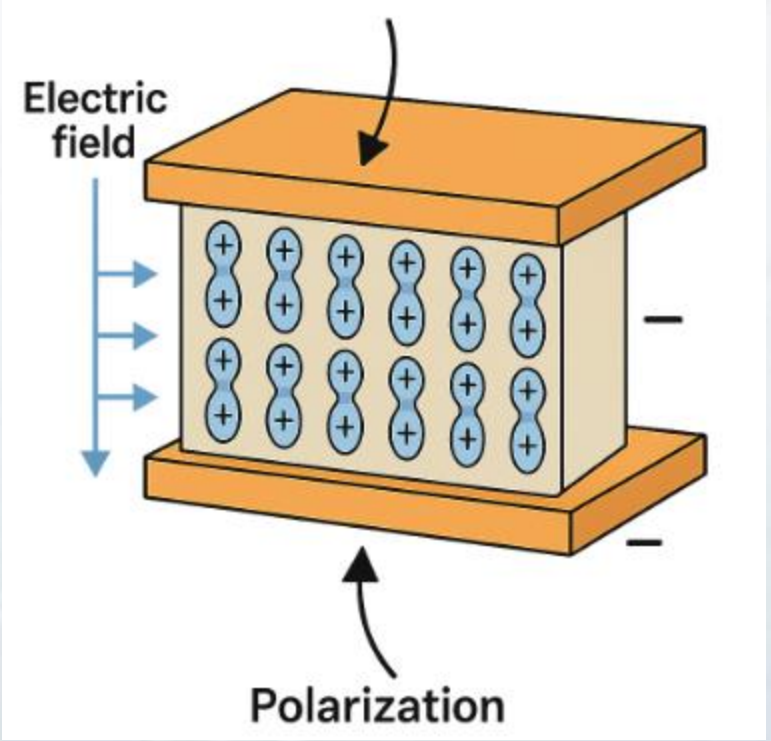
These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
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- Jeffrey Braunstein – ECE Department, University at Albany
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- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
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Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Electric Potential
- Electric Materials – Dielectrics
- Electric Materials – Conductors



https://www.ic-components.com/blog/capacitor-fundamentals-part-4-dielectric-polarization.jsp

Review

Steps to solve problems using Gauss's Law

1. Recognize the coordinate system.
2. Using symmetry, determine which components of the field exist.
3. Identify a Gaussian surface for which the sides are either parallel to or perpendicular to the field components. This surface is arbitrary in size.
4. Determine the total charge within that surface. The charges can be distributed on lines, surfaces or in volumes.

Review

5. To evaluate the electric flux passing through the Gaussian surface.
   - If the field is **parallel to the surface** $\oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = 0$
   - If the field is **constant and perpendicular to the surface** (surface area A),

$$\oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = D_i \oint \mathrm{d}A_i = D_i A_i$$

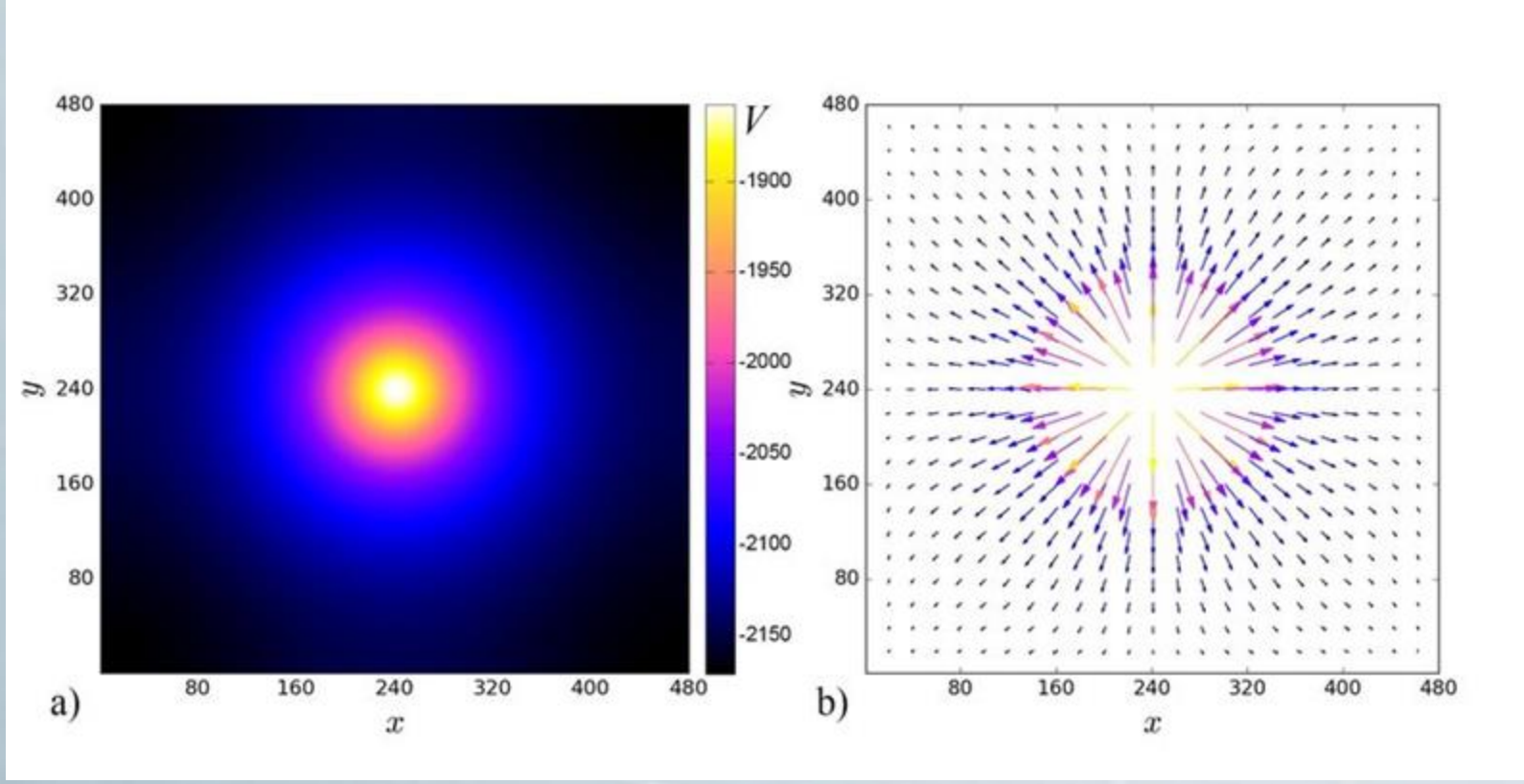
   where the subscript refers to the direction of the surface normal.

6. Now equate the two sides of Gauss' Law to find E:

$$D_i A_i = \oint \mathbf{D} \cdot \mathrm{d}\mathbf{S} = \int \rho \mathrm{d}V = Q_{enc}$$

Electric Potential

Electric Potential

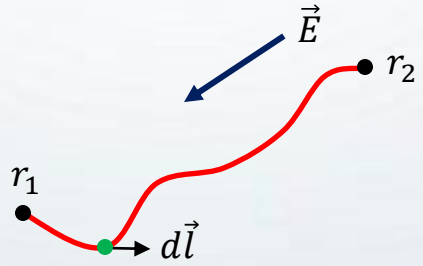


If the electric field looks like this (right), how do we derive the electric potential (left)? (Lauricella)

Electric Potential

- Circuits use electrical energy to do work
- How much work W is done by an electric field $\vec{E}$ that moves a single charge q along a path l?

$$W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{l} = \int_{r_1}^{r_2} q\vec{E} \cdot d\vec{l}$$



- This could be a complicated path integral

$$W = q\int_{t_1}^{t_2} \vec{E}\big(r(t)\big) \cdot d\vec{l}(t)$$

- Is there a simpler way?
- **Yes**! Let's look at a useful vector calculus identity for help

Electric Potential

- For any scalar field f: $\nabla\times(\nabla f) = 0$
- Looking at Maxwell's equations for electrostatics in differential form, we have:

$$\text{Gauss's law:} \quad \nabla \cdot \vec{D} = \rho$$

$$\text{Faraday's law:} \quad \nabla\times\vec{E} = 0$$

- This means that we can define some **scalar field V** such that $\vec{E} = -\nabla V$ which satisfies $\nabla\times(-\nabla V) = 0$
- V is called the **electrostatic potential**
- Our equation for work then simplifies to:

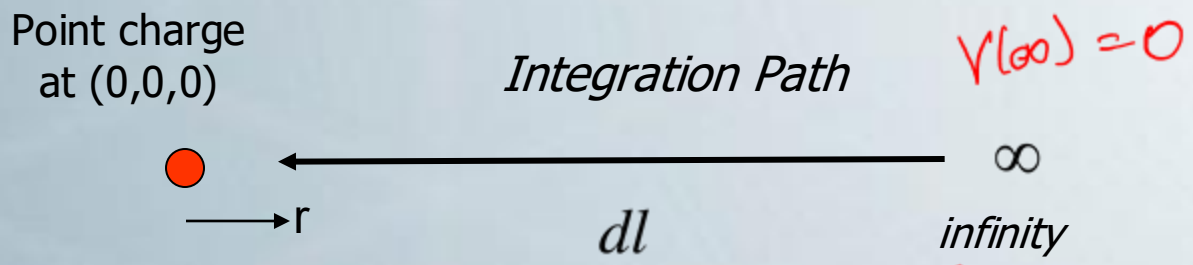
$$W = -q\int_{r_1}^{r_2}(\nabla V)\cdot d\vec{l} = -q[V(r_2) - V(r_1)] = -q\Delta V$$

Advantage #1: we no longer have to perform a path integral – W only depends on the end points, not the path the charge takes. In electrostatics $\vec{E}$ is a **conservative field**, due to $\nabla\times\vec{E} = 0$.

Advantage #2: we have a 2nd field we can use in electrostatics calculations and it is a **scalar field** → this is mathematically simpler to work with than a vector field

Electric Potential

- Example: Use case of point charge at origin and obtain the potential everywhere from the E-field



Spherical Symmetry

$$\vec{E} = \frac{q}{4\pi\epsilon_0 r^2}\hat{r}$$

Convenient reference if E field decreases away from source is $V(r_{ref} = \infty) = 0V$ (as $r \to \infty$)

Electric Potential

- The integral for computing the potential of the point charge is:

potential at point P = r

reference

$$V(r) - V(r=\infty) = -\int_{r=\infty}^{r} \vec{E} \cdot d\vec{l}$$

0

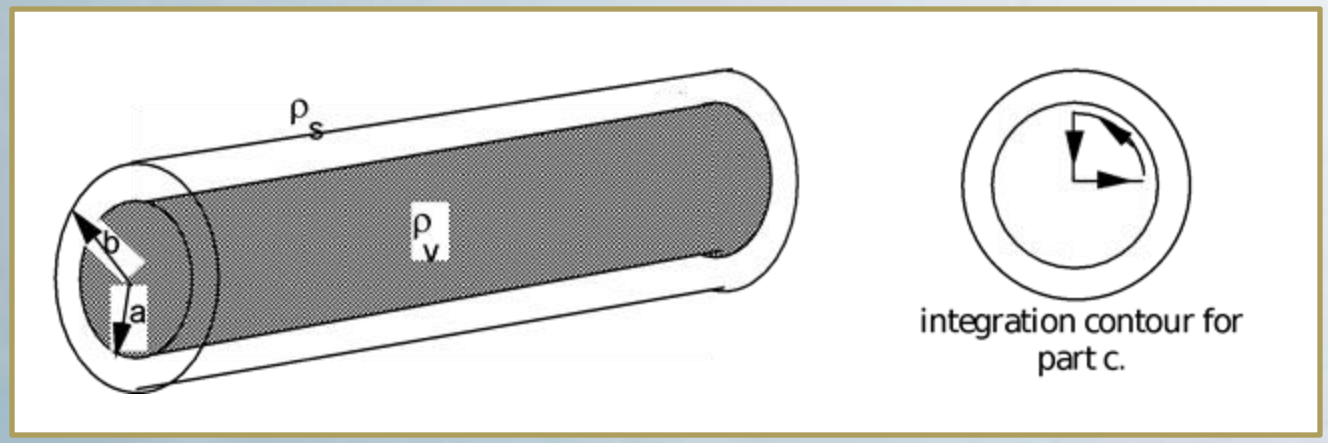
$$\therefore V(r) = -\int_{r=\infty}^{r} E \cdot dr'$$

Potential at a point charge a distance r away from the charge

$$= -\int_{r=\infty}^{r} \frac{q}{4\pi\varepsilon_0 r^2} \cdot dr \quad \Rightarrow \quad V(r) = \frac{q}{4\pi\varepsilon_0 r}$$

Electric Potential

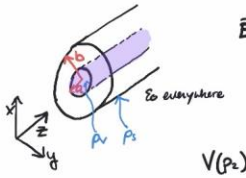
- What is the potential expression for this E-field from a shielded conductor with a grounded exterior?



$$E(r) = \begin{cases} \frac{\rho_v}{2\varepsilon_0} r\,\hat{r} & \text{for } 0 \le r < a \\ \frac{\rho_v a^2}{2\varepsilon_0}\frac{1}{r}\hat{r} & \text{for } a \le r < b \\ 0 & \text{for } r \ge b \end{cases}$$

Electric Potential

Potential of Shielded Cable



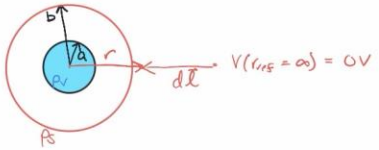
$$\vec{E}(\vec{r}) = \begin{cases} \frac{\rho_v r}{2\varepsilon_0}\hat{r} & r \le a \\ \frac{\rho_v a^2}{2\varepsilon_0 r}\hat{r} & a < r < b \\ 0 & r \ge b \end{cases}$$

$$V(\rho_2) - V(\rho_1) = -\int_{\rho_1}^{\rho_2} \vec{E}(\vec{r}) \cdot d\vec{l}$$

1) For $r > b$:

$$V(r) - V(\infty) = -\int_{\infty}^{r} \vec{E}\cdot d\vec{l} = -\int_{\infty}^{r} 0 \cdot dr\hat{r} = 0$$

$$V(r) = 0 \quad \text{for} \quad r \ge b$$

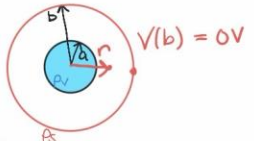


2) For $a \le r \le b$:

$$V(r) - V(b) = -\int_{b}^{r} \vec{E}\cdot d\vec{l} = -\int_{b}^{r} \left(\frac{\rho_v a^2}{2\varepsilon_0}\frac{1}{r'}\hat{r}\right)\cdot(\hat{r}\,dr')$$

$$= -\frac{\rho_v a^2}{2\varepsilon_0}\int_{b}^{r}\frac{1}{r'}dr' = -\frac{\rho_v a^2}{2\varepsilon_0}\ln(r/b) = \frac{\rho_v a^2}{2\varepsilon_0}\ln(b/r)$$

for $a \le r < b$

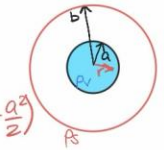


3) For $0 \le r < a$:

$$V(r) - V(a) = -\int_{a}^{r} \vec{E}\cdot d\vec{l} = -\int_{a}^{r} \left(\frac{\rho_v r'}{2\varepsilon_0}\hat{r}\right)\cdot(\hat{r}\,dr')$$

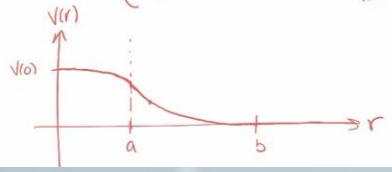
$$V(r) = V(a) - \frac{\rho_v}{2\varepsilon_0}\int_{a}^{r} r'dr' = \frac{\rho_v a^2}{2\varepsilon_0}\ln(b/a) - \frac{\rho_v}{2\varepsilon_0}\left(\frac{r^2}{2} - \frac{a^2}{2}\right)$$

$$= \frac{\rho_v a^2}{2\varepsilon_0}\ln(b/a) + \frac{\rho_v}{4\varepsilon_0}(a^2 - r^2) \quad \text{for} \quad 0 \le r < a$$



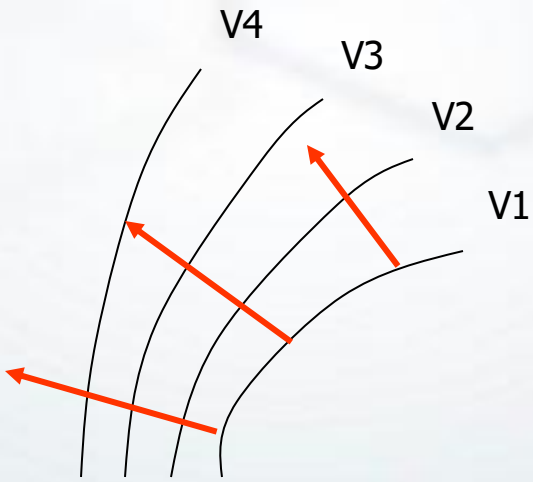
4) Full solution

$$V(r) = \begin{cases} \frac{\rho_v a^2}{2\varepsilon_0}\ln(b/a) + \frac{\rho_v}{4\varepsilon_0}(a^2 - r^2) & \text{for} \quad 0 \le r < a \\ \frac{\rho_v a^2}{2\varepsilon_0}\ln(b/r) & \text{for} \quad a \le r < b \\ 0 & \text{for} \quad r \ge b \end{cases}$$



Electric Potential

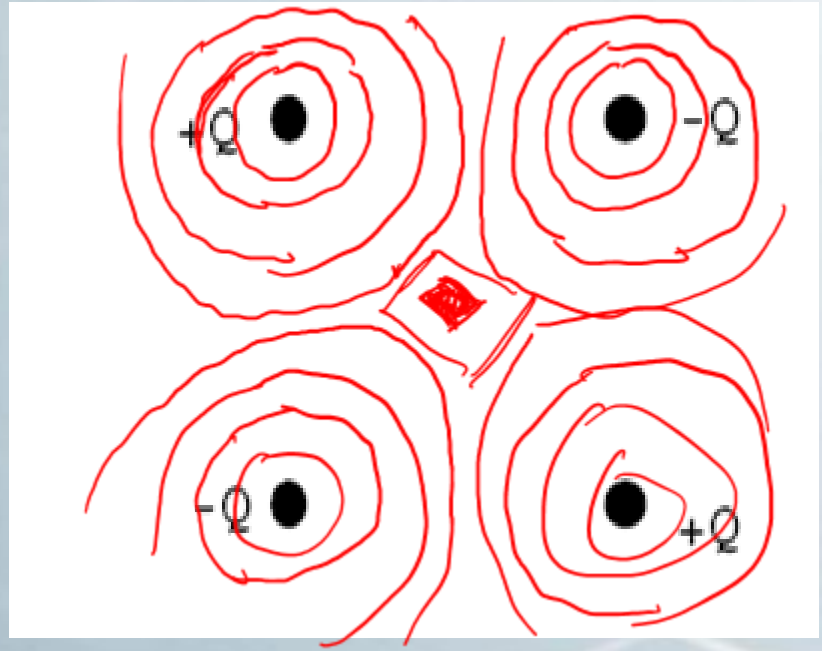
$$\vec{E} = -\nabla V$$

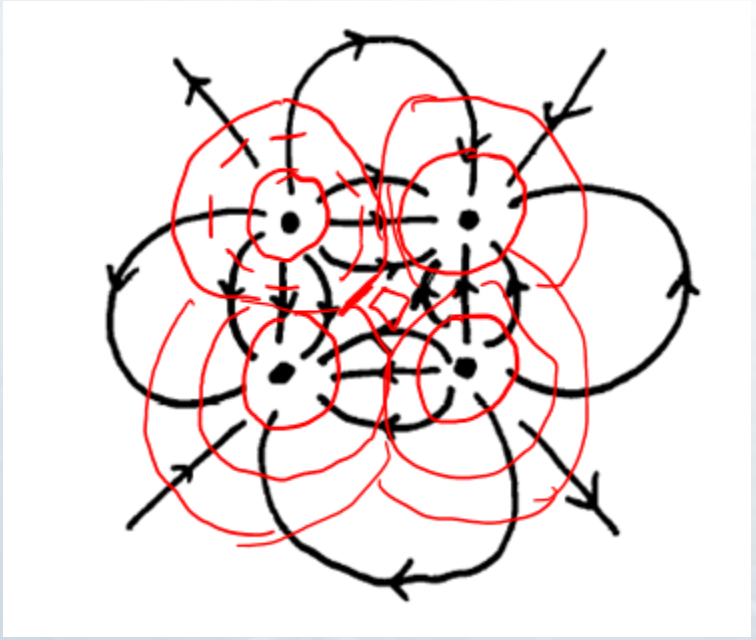


- Gradient points in the direction of largest change
- Therefore, E-field lines are perpendicular (normal) to constant V surfaces
- In terms of elevation, lines of constant V are lines of constant elevation and the E field points in the direction of steepest descent

Electric Potential

- Plot a set of equipotentials for this quadrupole.





E-field

Electric Potential

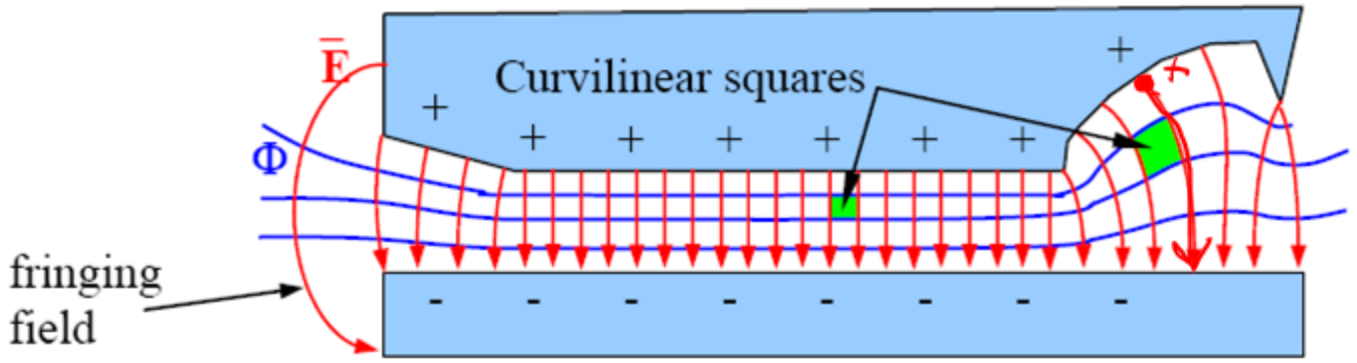


Figure R11-3. Graphical field mapping of $\bar{E}$ and F between charged conductors

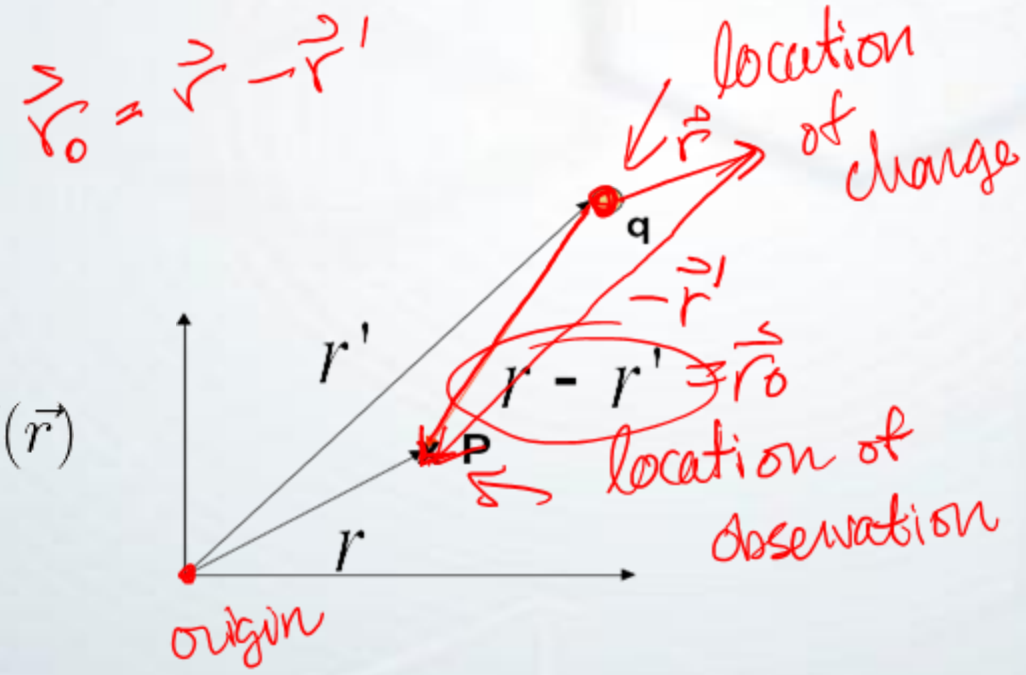
http://ocw.mit.edu/NR/rdonlyres/Electrical-Engineering-and-Computer-Science/6-013Electromagnetics-and-ApplicationsFall2002/922D1A06-9AC9-4076-B1F5-066EE896043C/0/Rec11Notes.pdf

Electric Potential

Potential of a single charge

- For the case of a point charge:

$$V = \frac{q}{4\pi\epsilon_0 r} = \frac{q}{4\pi\epsilon_0 (r - r')} = V(\vec{r})$$



$\vec{r}$ is the field point where we are measuring/calculating V (observation point)

$\vec{r}'$ is the location of charge contributing to the potential

Electric Potential

- For a charge distribution:

Potential at observation point r

Charge density at a location r′

$$V(r) = \int\int\int \frac{\rho(r') \cdot dv'}{4\pi\epsilon_0 |r - r'|}$$

Volume charge distribution

Result: integral over volume, adding up the potential contributed by charge density at each point r′

Distance between r′ and observation point r

$$V(r) = \int \frac{\rho(r') \cdot dl'}{4\pi\epsilon_0 |r - r'|}$$

Line charge distribution

Result: integral over line, adding up the potential contributed by charge density at each point r′ along the line dl′

Electric Materials

Electric Materials

- Previously we said:

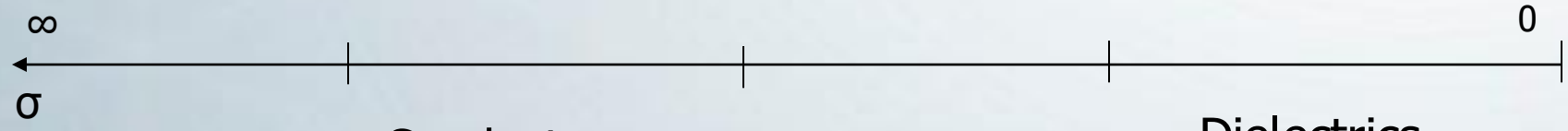
$$\oint \vec{D} \cdot ds = \int \rho \cdot dv$$

$$\vec{D} = \epsilon \vec{E}$$

- What about situations in which $\boldsymbol{\varepsilon} = \epsilon_r \epsilon_0 \neq \boldsymbol{\varepsilon}_0$? relative permittivity

- Materials other than free space can have a wide range of $\boldsymbol{\varepsilon}$.

Electric Materials



Superconductors

$\sigma \sim \infty$

Semiconductors

mid σ's

σ (for Germanium) ~ 2.2 S/m

Perfect dielectric

Perfect Conductor

Conductors

High conductivities;
σ (for Copper) ~ 5.8×10^{7} S/m

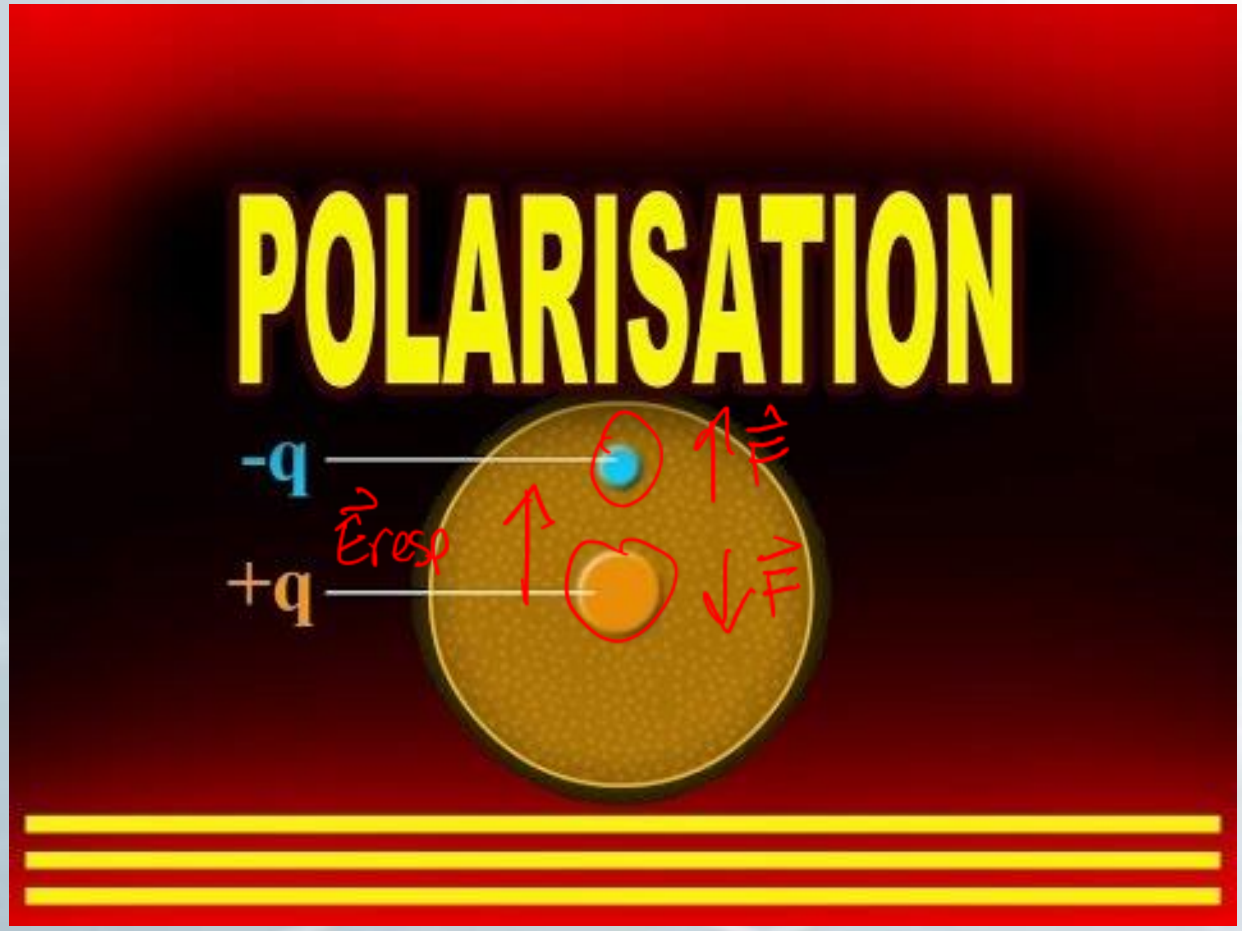
Dielectrics

Low conductivities;
σ (for Rubber) ~ 1×10^{-15} S/m

Permittivities, $\varepsilon = 1\text{-}100\varepsilon_0$

Note: ε_0 is the permittivity of free space / vacuum = 8.854×10^{-12} F/m

Electric Materials



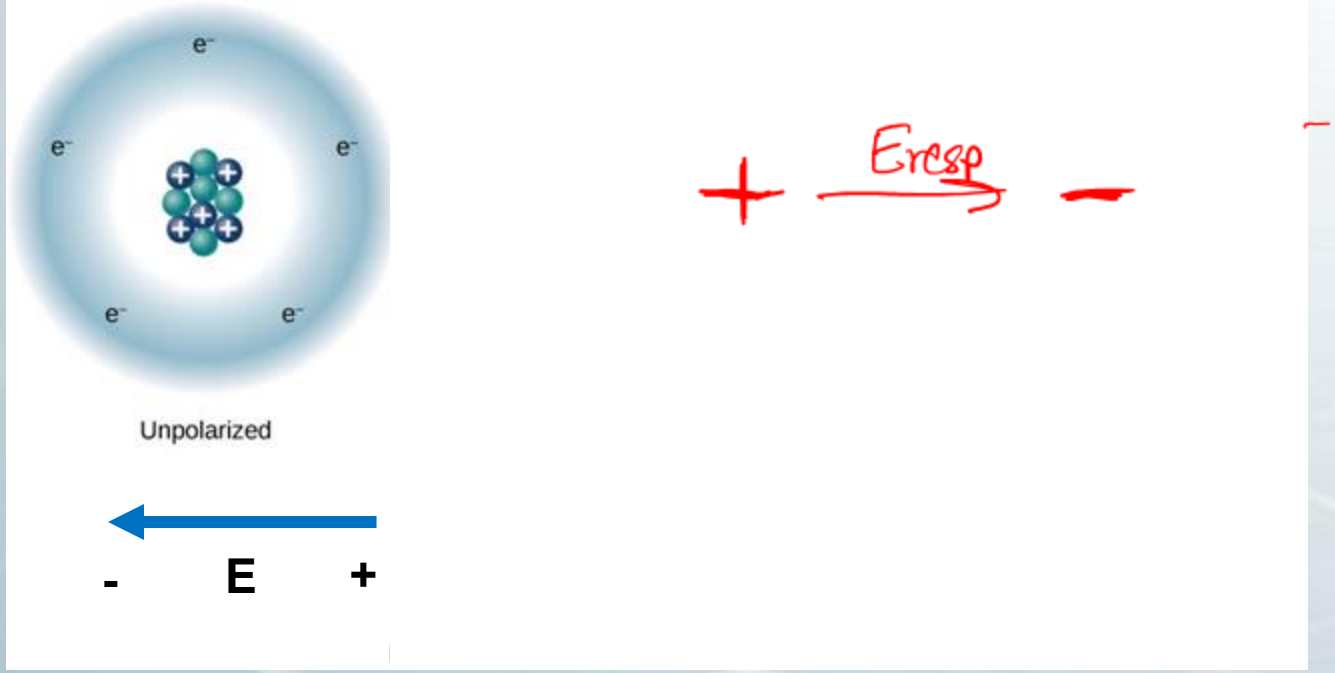
Physics4Students

$\vec{F} = q\vec{E}$

$\vec{E}_{ext}$

Electric Materials

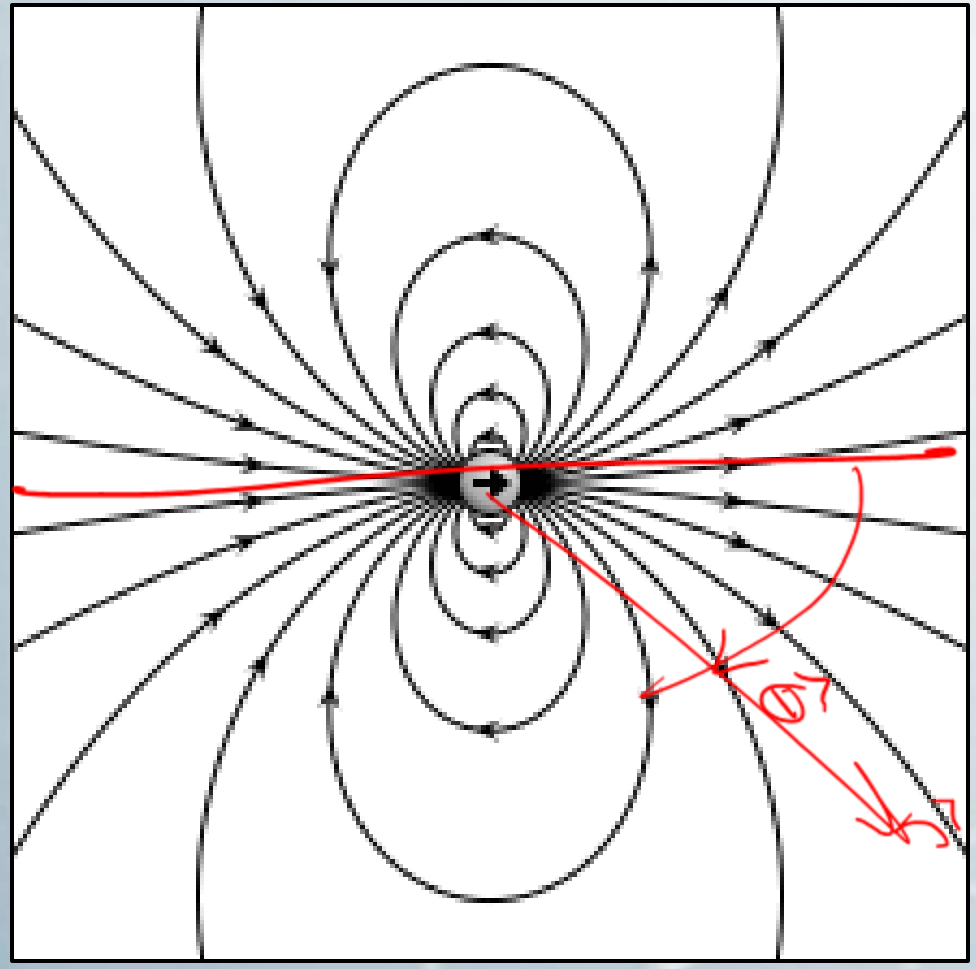
- When an electric field is applied to a dielectric:
  - Electrons (bound to their positively-charged cores) move towards the + side of the **E** field
  - The positive core moves towards the – side of the **E** field



This forms many **dipoles** in the material

Electric Materials

Dipoles

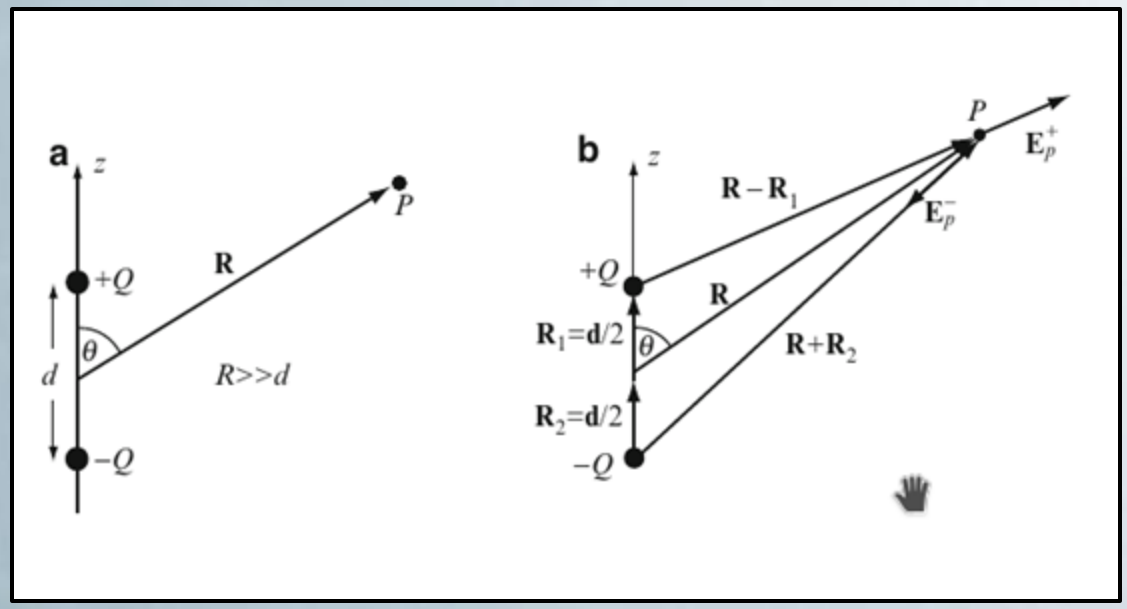


Wikipedia

- When two opposite charges are very close together relative to some much larger distance d from which we measure their field, we call them an electric dipole.

Electric Materials

Dipoles



$$\mathbf{E}_d = \mathbf{E}_p^+ + \mathbf{E}_p^- = \frac{Q}{4\pi\varepsilon_0}\left(\frac{(\mathbf{R}-\mathbf{d}/2)}{|\mathbf{R}-\mathbf{d}/2|^3} - \frac{(\mathbf{R}+\mathbf{d}/2)}{|\mathbf{R}+\mathbf{d}/2|^3}\right) \quad \left[\frac{\mathrm{N}}{\mathrm{C}}\right]$$

Ida

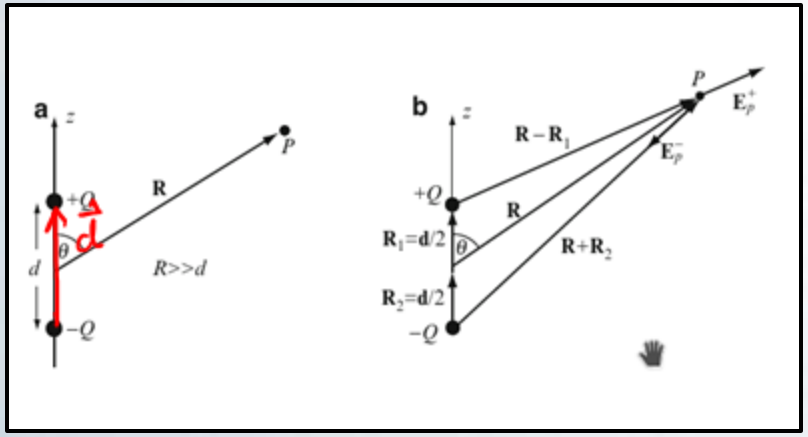
- Superposition allows us to write an expression for the electric field caused by two opposite-value point charges, separated by distance d, at a distance R away.

Note: here, we will use bold letters to represent vectors.

Which coordinate system is this?

Electric Materials

Dipoles



… after some algebra, infinite series math and limits, we can write

$$\mathbf{E}_d \approx \frac{1}{4\pi\varepsilon_0 R^3}\left[3\frac{Rp\cos\theta}{R^2}\hat{\mathbf{R}}R - p\left(\hat{\mathbf{R}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta\right)\right] = \frac{p}{4\pi\varepsilon_0 R^3}\left(\hat{\mathbf{R}}2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta\right) \quad [\mathrm{N/C}]$$

We let $\mathbf{p} = Q\mathbf{d}$ and call $\mathbf{p}$ the dipole moment.

$\mathbf{d}$ points from the negative charge to the positive.

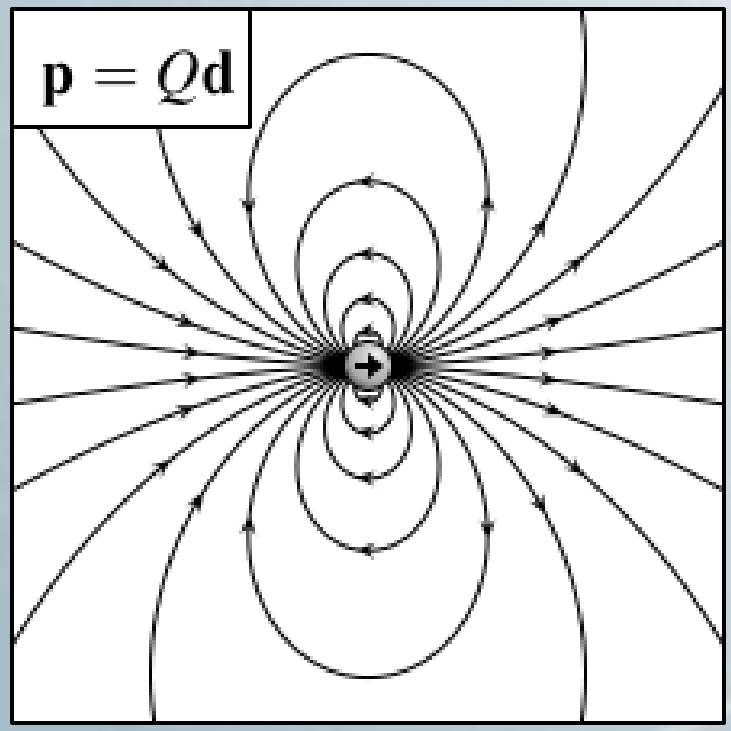
points opposite to $\vec{E}$ field from dipole

$\vec{E}$ " " " pos. " " " negative

Electric Materials

Dipoles

$$\frac{p}{4\pi\varepsilon_0 R^3}\left(\hat{\mathbf{R}}2\cos\theta + \hat{\boldsymbol{\theta}}\sin\theta\right) \quad [\mathrm{N/C}]$$



Wikipedia

A few things to note:

- This is an inverse cube expression instead of the standard inverse square expression of Coulomb's Law
- Field intensity is maximum at points aligned with the dipole and minimum perpendicular to the dipole
- Field intensity is proportional the distance between the charges
- Anything that causes the charges to move apart will proportionally increase their dipole field

Electric Materials

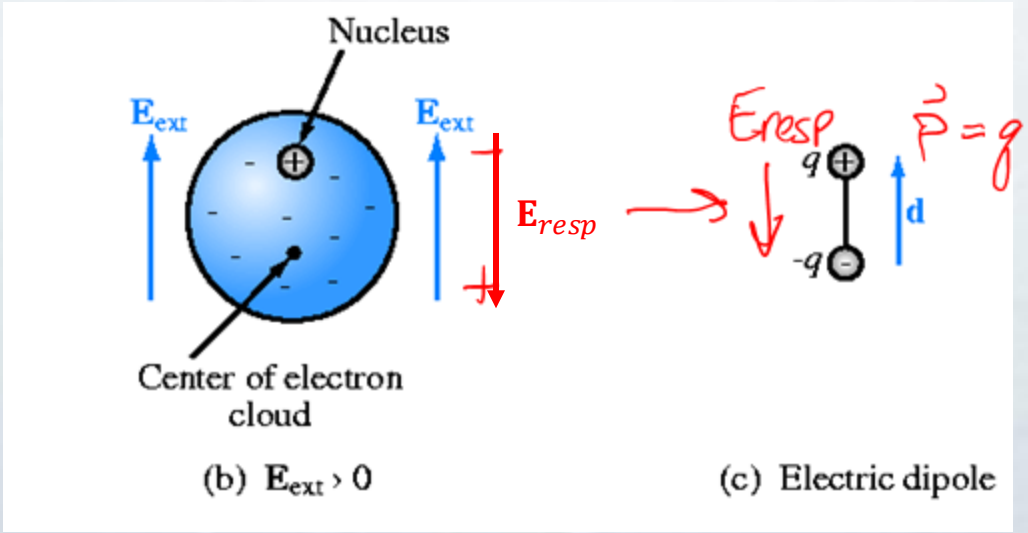
$$\vec{D} = \epsilon \vec{E}$$

Define: $\vec{p} = q\vec{d}$ = dipole moment

$E_{response} \propto -\sum p_i = P$ ⟵ Polarization

$\boxed{E_{response}}$ partially cancels applied Field

$$\therefore E_{total} = E_{external} + E_{response} \neq 0$$



↓ less than E_{ext} → polarization partially cancels the applied field in the dielectric

Electric Materials

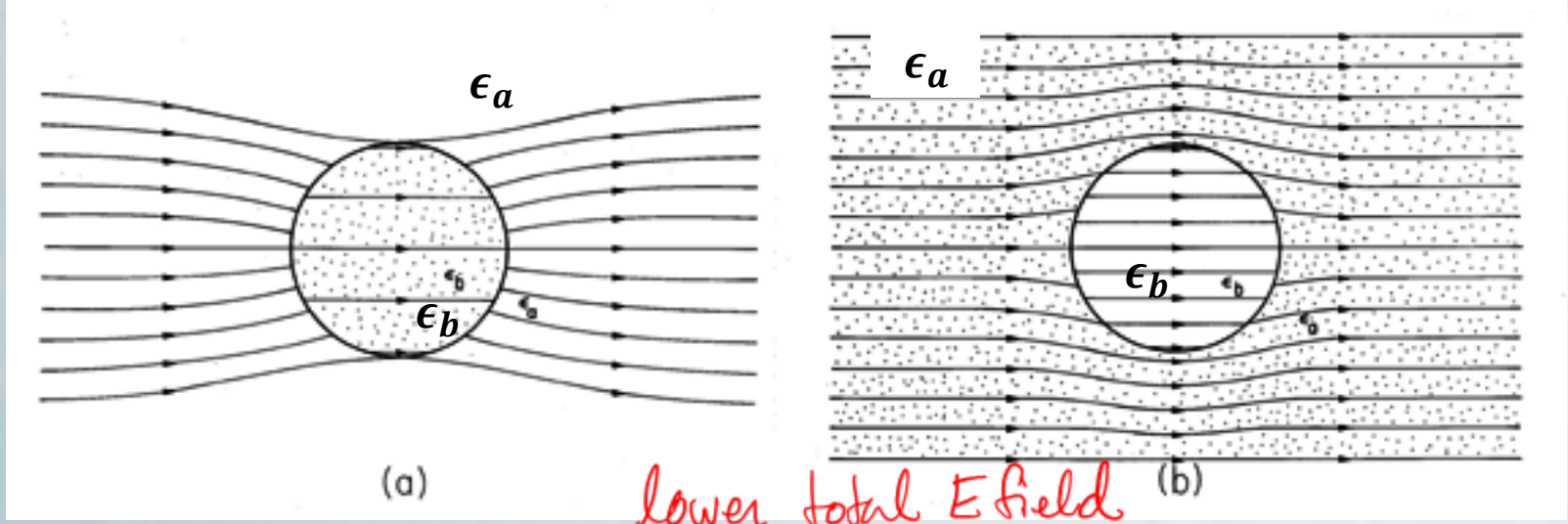
- Different materials have different permittivities
- Permittivities will also depend on field frequency, temperature, humidity, and other factors

Material	Relative Permittivity (ϵ_r)
Paper	1.4
Concrete	4.5
Silicon	11.68
Water	50-90
Titanium dioxide	86-173
Calcium copper titanate	>250,000

Wikipedia

Electric Materials

StackExchange



lower total E field
more polarization

a.) $\varepsilon_b > \varepsilon_a$ permittivity for sphere > permittivity outside

b.) $\varepsilon_b < \varepsilon_a$ " " " < " "

less polarization

Note the effect on relative E-field density.

larger total E field

Do Lecture 11 Exercise 1 in groups of up to 4 on Gradescope

Electric Materials - Dielectrics

Dielectrics

Gauss' Law uses just free charge

$$\oint \vec{D} \cdot d\vec{s} = \int \rho dV = Q_{enc}$$

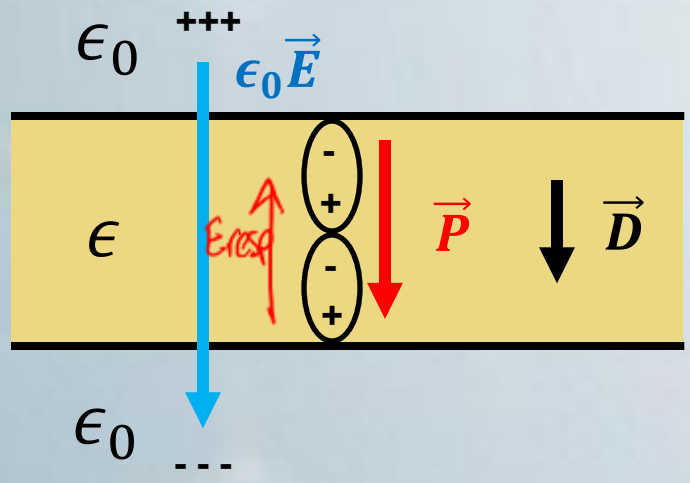
Examples of free charges:

- electrons in a conductor
- ρ_s on conductor
- electron beam
- electrons or holes in a doped region of semi-conductor

- But in dielectrics, we also encounter something called **bound charge**

Dielectrics

- Consider a parallel plate capacitor with a material of electric permittivity ϵ inside and ϵ_0 outside.



1. We apply an electric field $\vec{E}$ across the plates.

2. Polarization causes the formation of a *polarization density* field $\vec{P}$ that opposes $\vec{E}$

3. The electric displacement field $\vec{D}$ arises due to the remaining net charge density in the dielectric

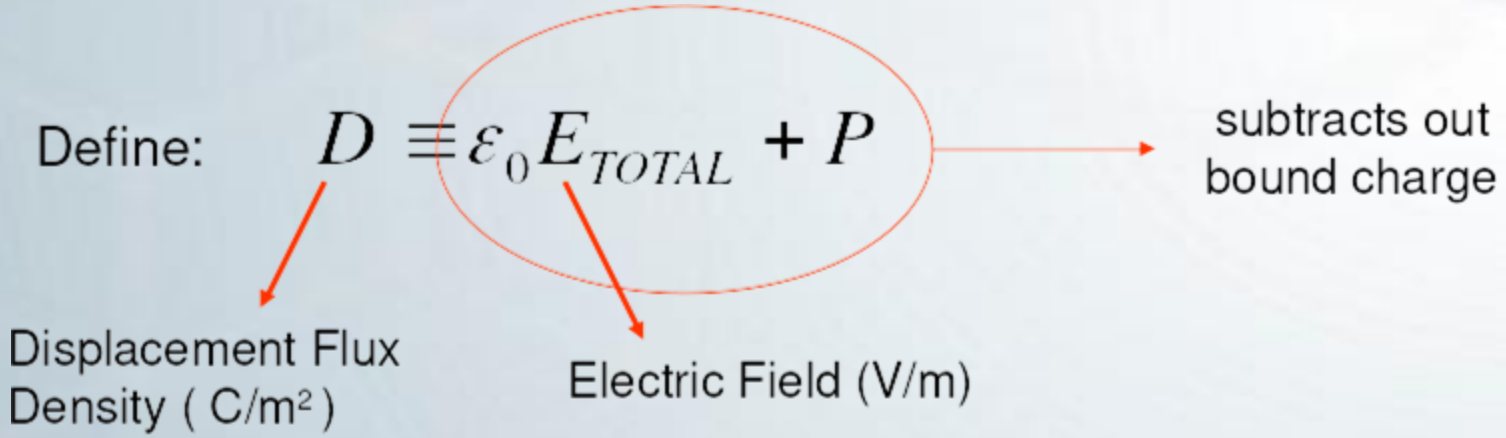
$$\epsilon_0\vec{E} + \vec{P} = \vec{D}$$

$$\rho_f + \rho_b = \epsilon_0\vec{\nabla}\cdot\vec{E}$$

$$-\vec{\nabla}\cdot\vec{P}$$

$$\vec{\nabla}\cdot\left(\epsilon_0\vec{E} + \vec{P}\right) = \vec{\nabla}\cdot\vec{D} = \rho_f$$

Dielectrics



E_{TOTAL} is due to bound/dielectric charge and free charge

P is due to bound/dielectric charge only and opposite sign

D is due to free charge only

Watch until ~6 mins: https://www.youtube.com/watch?v=rHO4JdIiwFg&ab_channel=DrBenYelverton

Dielectrics

- If the electric field intensity is high enough in a dielectric, electrons will eventually become unbound from their atoms
- This creates a conductive path through the material, which can become permanent if it changes the material composition
- This is known as **dielectric breakdown** and is usually bad, since dielectrics are used as electric insulators

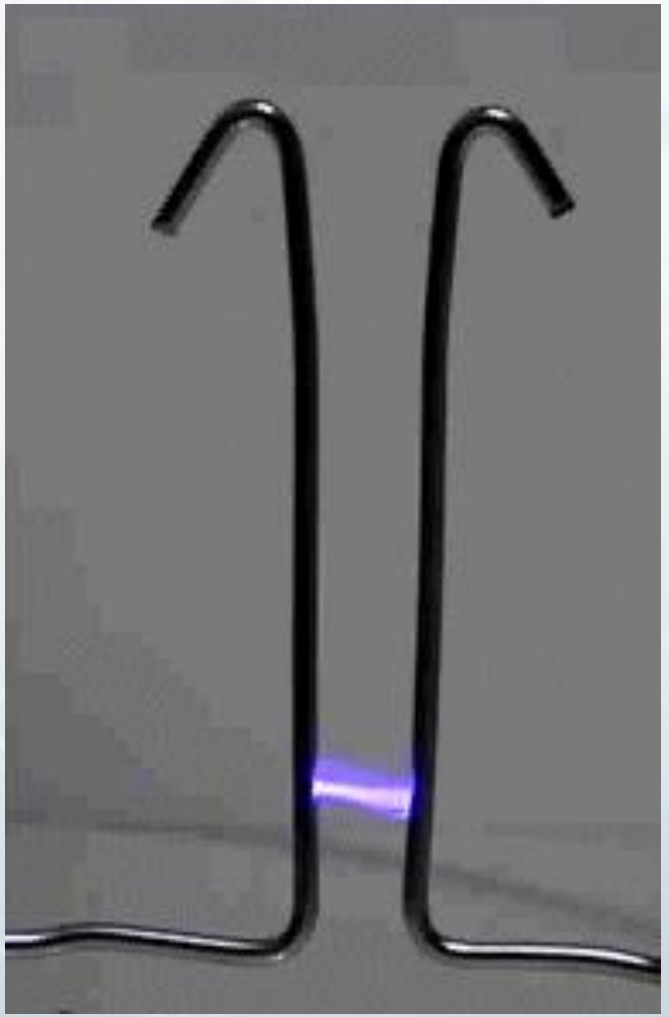
Dielectrics

- A dielectric will break down if

$$\vec{E}_{applied} > \vec{E}_{d-strength}$$

- As electrons begin to flow, they collide with and dislodge more electrons, leading to more current in an **avalanche process**.

- For two cables separated by an air gap, this breakdown field value is 30kV/cm.



"Jacob's Ladder" (Gifer)

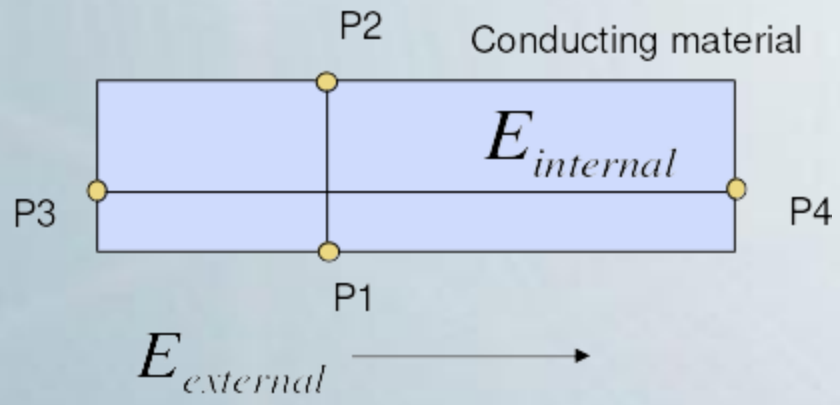
Electric Materials - Conductors

Conductors

- Why aren't any metals listed in the table of dielectric constants?
- Because metals aren't dielectrics; they're conductors
- In dielectrics, E-fields cause electrons and nuclei to "flex" and form dipoles, but the electrons generally **remain bound to the nuclei**.
- In conductors, that isn't the case - a large number of electrons are **free to move around under the influence of a field**.

Conductors

- The inside of a perfect conductor is an **equipotential** region, which means that the potential is the same everywhere.



$$V(P_2) = V(P_1) = V(P_3) = V(P_4)$$

$$V(P_2) - V(P_1) = V(P_3) - V(P_4) = 0$$

$$= -\int_{P_1}^{P_2} E_{internal} \bullet dl \Rightarrow E_{internal} = 0$$

- What are the consequences of this?

- Therefore, **in a perfect conductor,** $\vec{E} = 0$

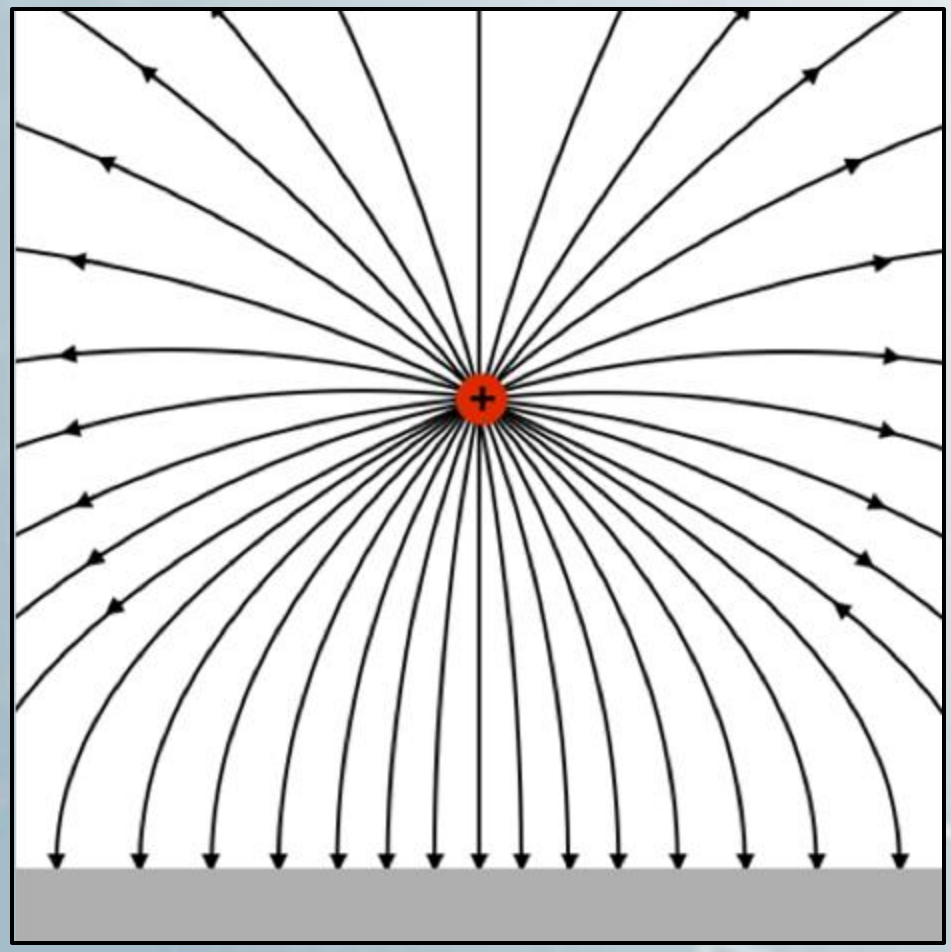
Conductors

- Let's return to the permittivity ε. What can we conclude about its value in perfect conductors now?

$$\vec{D} = \epsilon \vec{E} \longrightarrow \epsilon = \frac{|\vec{D}|}{|\vec{E}|}$$

- Since the E-field is zero in an ideal conductor, $\boldsymbol{\varepsilon}$ is effectively infinite!
- Hence, permittivity isn't a meaningful measurement for conductors

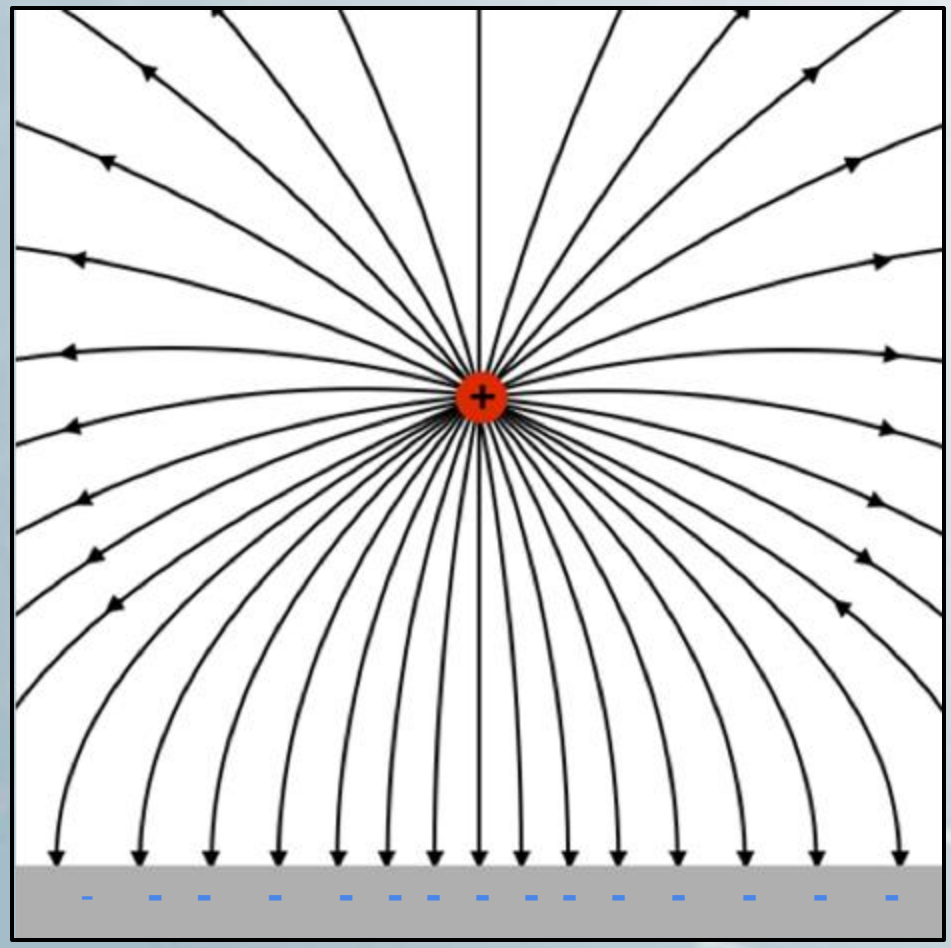
Conductors



Physics LibreTexts

- Previously we looked at a charge suspended over an infinite plane.
- What if that plane is a conductor? How will the plane behave in this scenario?

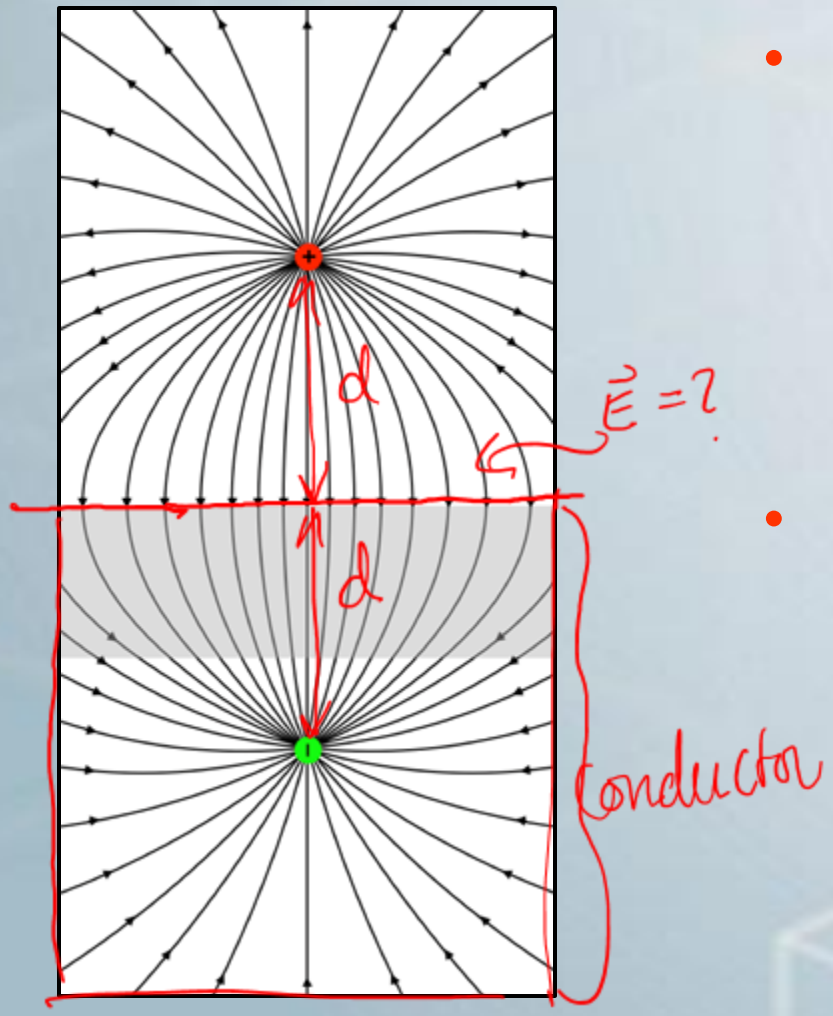
Conductors



Physics LibreTexts

- The plane will accumulate charge near the surface that **completely** negates the E-field of the (+) charge.
- How do we draw field lines for a situation like this? We need to take all charges and charge distributions into account

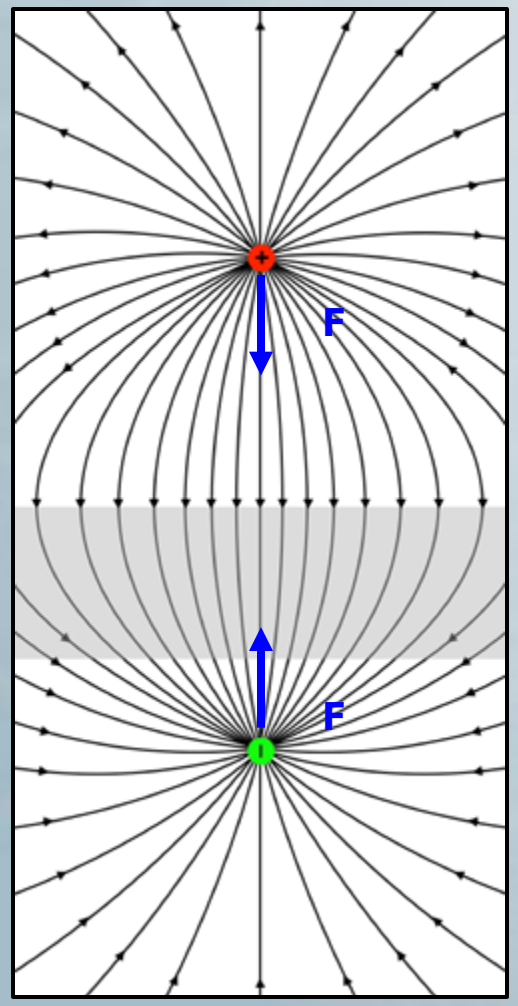
Conductors



Physics LibreTexts

- The conductor's surface charge influences the field in the exact same way as if we removed the conductive plan and added a "mirror image" of the (+) charge at the same distance from the surface and with opposite charge.
- We can use this to our advantage to solve for electric fields near perfect conductors

Conductors



Physics LibreTexts

- All fields and forces will be the same as if this were the case above the plane, and this works for distributions of charge as well.
- When solving problems this way, we call it the "**method of images**".
  1. Add equal and opposite charge distributions an equal distance from the conductor surface
  2. Calculate the E-field from each charge distribution
  3. Use superposition to find the E-field in the region outside the conductor (inside the conductor it's 0)

Do Lecture 11 Exercise 2 in groups of up to 4 on Gradescope

Wrap-Up

- HW5 posted yesterday